Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	27 June 2026
Team ID	
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	4 Marks

Technical Architecture:

The proposed Human Development Index (HDI) Predictor follows a three-layer architecture consisting of the User Interface Layer, Flask Backend Layer, and Machine Learning Layer. The application accepts socio-economic indicators from users, preprocesses the data, predicts the HDI score using a trained Linear Regression model, classifies the result into Very High, High, Medium, or Low HDI, and displays the prediction through a Flask web interface.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface for entering HDI indicators and displaying prediction results	HTML5, CSS3, Bootstrap, JavaScript
2	Application Logic-1	Request handling, input validation, routing	Python, Flask
3	Application Logic-2	Data preprocessing, feature engineering, data cleaning	Pandas, NumPy, Scikit-learn
4	Application Logic-3	HDI prediction engine	Linear Regression (Scikit-learn)
5	Dataset Storage	Store HDI dataset (CSV)	CSV File
6	Model Storage	Store trained ML model	Pickle (.pkl)
7	File Storage	Store dataset, trained model, graphs, reports	Local File System
8	Visualization	Generate charts and statistical graphs	Matplotlib, Seaborn
9	Machine Learning Model	Predict Human Development Index	Linear Regression
10	Prediction Engine	Predict HDI score and classify development level	Scikit-learn
11	Infrastructure (Server / Cloud)	Application deployment and hosting	Flask Server, IBM Cloud / Render / Railway

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks and libraries used in the project	Flask, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
2	Security Implementations	Input validation, secure request handling	Flask Validation, HTTPS
3	Scalable Architecture	Three-layer architecture separating UI, backend, and ML model	3-Tier Architecture, Flask
4	Availability	Accessible through any modern web browser	Flask Server, IBM Cloud / Render
5	Performance	Fast prediction using optimized preprocessing and trained model	Pandas, NumPy, Linear Regression

Technical Architecture Description: